MASS Syndrome (MASS Phenotype): Comprehensive Disease Characteristics Report

Causal gene: FBN1 (fibrillin-1, HGNC:3603, 15q21.1). MASS syndrome is a heritable connective-tissue disorder caused by heterozygous pathogenic variants in **FBN1**, the same gene responsible for Marfan syndrome. The acronym **MASS** denotes **M**itral valve prolapse, **A**ortic enlargement (borderline, non-progressive), and **S**kin and **S**keletal findings. It occupies a defined point on the phenotypic continuum of *type-1 fibrillinopathies*, distinguished from classic Marfan syndrome principally by the **absence of ectopia lentis** and the **absence of a progressive aortic root aneurysm**. Identifiers: **OMIM 604308**, **MONDO:0011431**.

> *Note on scope and evidence:* No patient-level dataset was provided; this report is synthesized from aggregated disease-level resources and primary literature (PubMed). Because MASS and Marfan syndrome are **allelic FBN1 disorders on one continuum**, much mechanistic, therapeutic, and model-organism evidence is derived from Marfan syndrome / FBN1 studies and is explicitly flagged as such. Evidence types are labeled: *human clinical*, *model organism*, *in vitro*, *computational/consensus*.

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1. Disease Information

Overview. MASS phenotype was first delineated by **Glesby & Pyeritz (JAMA 1989,** **),** P 2739055 who observed that more than half of patients referred for a possible heritable connective-tissue disorder could not be classified in the existing nosology yet shared Marfan-like systemic features. They proposed the acronym MASS "to emphasize involvement of the mitral valve, aorta, skeleton, and skin," framing mitral valve prolapse (MVP) patients as "a continuum, from Marfan syndrome at one extreme to isolated mitral valve prolapse due to myxomatous proliferation of the valve leaflets" (*human clinical*,

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P 2739055

MASS is thus an **overlap connective-tissue disorder**: patients have mild aortic root dilatation and MVP with systemic (skin/skeletal) findings, but do **not** meet full criteria for Marfan syndrome and lack lens dislocation.

Key identifiers. - **OMIM:** 604308 (MASS SYNDROME; MASS PHENOTYPE) - **MONDO:** MONDO:0011431 - **Orphanet:** MASS falls within the Marfan/type-1-fibrillinopathy spectrum (ORPHA:284963 "Marfanoid–overlap" spectrum; no unique high-level ORPHA code is universally applied — *data incompletely standardized*) - **ICD-10:** Q87.4 relates to Marfan syndrome; MASS has no distinct ICD-10 code and is usually coded under connective-tissue/MVP categories (I34.1 mitral valve prolapse; Q87.4) — *not uniquely coded* - **ICD-11:** LD28.0 (Marfan syndrome) / connective-tissue categories — *not uniquely coded* - **MeSH:** No dedicated MeSH term; indexed under "Marfan Syndrome" / "Mitral Valve Prolapse" / "Connective Tissue Diseases" - **Gene:** FBN1 — HGNC:3603, NCBI Gene 2200, Ensembl ENSG00000166147, UniProt P35555; gene-level OMIM *134797*

Synonyms / alternative names. MASS phenotype; MASS spectrum; mitral valve–aorta–skeleton–skin syndrome. (Historically grouped with "marfanoid" and "overlap heritable connective-tissue disorder.")

Information source. Disease-level / aggregated resources and primary literature (not EHR-derived).

Ontology: MONDO:0011431; disease class — Mendelian connective-tissue disorder.

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2. Etiology

Primary cause — genetic. MASS is a **monogenic, autosomal-dominant** disorder caused by heterozygous pathogenic variants in **FBN1**. There is no infectious or primary environmental cause. FBN1 "encodes fibrillin-1, a large glycoprotein which is a major component of microfibrils that are widely distributed in structural elements of elastic and non-elastic tissues" (*human clinical/review*,

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P 26875674

The same review groups MASS among the "type-1 fibrillinopathies," which "include Marfan syndrome (MFS), MASS syndrome … [and] Familial ectopia lentis."

Genetic risk factors. - *Causal variants:* heterozygous FBN1 variants (see §4). The pathogenic FBN1 allele is both necessary and, with age-dependent penetrance, sufficient. - *Modifier genes / genetic background:* variable expressivity within and between families implies modifier effects; specific robust MASS modifier loci are **not established** (*data not available*). Candidate modifiers in the broader fibrillinopathy literature include genes of the TGF-β axis. - *Family history* is the dominant risk factor (50% transmission risk from an affected parent).

Environmental risk factors. None cause MASS. However, **hemodynamic/mechanical stress** (hypertension, isometric/heavy resistance exercise, pregnancy) is an aggravating factor for the aortic and valvular components across the FBN1 spectrum; pregnancy increases cardiovascular risk and aortic events in FBN1 disorders (*human clinical,*

; P 42880760

). P 20351702

Age and (for aortic outcomes) male sex modulate cardiovascular expression.

Protective factors. No genetic protective alleles are defined for MASS. Pharmacologic "protection" against aortic growth (beta-blockers, ARBs) and lifestyle modification (avoidance of competitive/isometric sport, blood-pressure control) reduce cardiovascular risk (*human clinical,*

). P 30261963

Gene–environment interactions. The FBN1 lesion sets a structurally weakened, TGF-β–dysregulated extracellular matrix; superimposed mechanical wall stress then drives the cardiovascular phenotype. This gene × hemodynamic-load interaction underlies the rationale for mechanical-stress-reducing therapy (*model organism / human clinical,*

; P 17892982

). P 41195532

3. Phenotypes

MASS is defined by a **tetrad** of organ systems with **variable expressivity** and generally **mild-to-moderate** severity. By definition the aortic component is borderline and the lens is normal. Onset is typically **childhood to young adulthood**; most features are slowly **progressive or stable** rather than episodic.

Phenotype	Type	HPO term	Onset / course	Frequency (qualitative)
Mitral valve prolapse (± mitral regurgitation)	Clinical sign (cardiac)	HP:0001634 / HP:0001653	Childhood–adult; may progress	Core/ common
Borderline aortic root dilatation (Z-score <2, non-progressive)	Clinical sign (cardiac)	HP:0002616 (aortic root aneurysm, mild)	Young adult; **non-progressive by definition**	Core/ common
Striae atrophicae (skin striae)	Physical manifestation (skin)	HP:0001065	Adolescence+	Common
Skin hyperextensibility / thin skin	Physical manifestation (skin)	HP:0007429 / HP:0000974	Childhood	Variable
Tall/long limbs, dolichostenomelia	Physical manifestation (skeletal)	HP:0001519 / HP:0001166 (arachnodactyly)	Childhood	Variable
Pectus excavatum / carinatum	Physical manifestation (skeletal)	HP:0000767 / HP:0000768	Childhood–adolescence	Common
Scoliosis	Clinical sign (skeletal)	HP:0002650	Childhood–adolescence	Variable
Joint hypermobility	Clinical sign (skeletal)	HP:0001382	Congenital/ childhood	Common
Pes planus (flat feet)	Physical manifestation	HP:0001763	Childhood	Common
High-arched palate	Physical manifestation	HP:0000218	Congenital	Variable
Myopia (mild, **no ectopia lentis**)	Clinical sign (ocular)	HP:0000545	Childhood	Variable

Explicitly ABSENT / excluded (key discriminators): ectopia lentis (HP:0001083) and progressive aortic aneurysm/dissection — their presence reclassifies the patient toward Marfan syndrome (*human clinical,* ; P 20591885). P 2739055

Severity / progression. Generally mild and non-life-threatening; skeletal and skin features are static or slowly progressive. MVP can progress to significant mitral regurgitation in a minority. Because "the clinical phenotype … constitutes a continuum" (P 2739055), individual severity varies widely.

Quality-of-life impact. Usually modest: musculoskeletal pain, fatigue, joint instability, and body-image concerns from skeletal/skin features; palpitations/arrhythmia and exertional limitation when MVP/regurgitation is significant. No MASS-specific EQ-5D/SF-36 datasets exist (*data not available*); QoL inference is drawn from the broader MVP/connective-tissue literature.

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4. Genetic / Molecular Information

Causal gene. FBN1 — fibrillin-1; HGNC:3603; NCBI Gene 2200; Ensembl ENSG00000166147; UniProt **P35555**; cytogenetic locus **15q21.1**; gene OMIM 134797; 65–66 coding exons (P 26875674).

Pathogenic variants. - *Gene(s) affected:* FBN1 (single gene). - *Variant classification (ACMG/AMP):* pathogenic / likely pathogenic FBN1 variants; a large fraction of rare FBN1 missense changes remain VUS, complicating MASS (a mild-phenotype) diagnosis. Interpretation uses ClinVar/ClinGen and the FBN1-specific literature. - *Variant type/class:* predominantly **missense**, classically **cysteine substitutions within calcium-binding EGF-like (cbEGF) domains** that disrupt disulfide bonding and calcium binding (the same class modeled by the mouse *Fbn1^C1039G* allele; §15). Also nonsense, frameshift, splice-site, and in-frame deletions. Sibling cases with isolated skeletal Marfan features and no cardiac/ocular findings carried the recurrent missense **R2726W (p.Arg2726Trp)**, illustrating how particular FBN1 alleles produce

mild/partial phenotypes (*human clinical*,

P 26875674

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- *Allele frequency:* pathogenic FBN1 variants are individually **ultra-rare** in gnomAD (most absent or singletons); FBN1 is constrained against loss-of-function. - *Somatic vs germline:* **germline** (constitutional); de novo variants occur (as in Marfan, ~25% de novo). - *Functional consequence:* two principal mechanisms — (i) **dominant-negative** (mutant fibrillin-1 monomers poison microfibril assembly), typical of cbEGF cysteine missense; and (ii) **haploinsufficiency** (premature-termination/null alleles reducing fibrillin-1 quantity). Both converge on impaired microfibril function and dysregulated TGF-β sequestration (*human clinical/review*,

P 26919284

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Genotype–phenotype note. No FBN1 variant is pathognomonic for MASS specifically; MASS reflects the *milder end* of FBN1 expression, and the same variant can produce Marfan syndrome, MASS, or isolated features in different individuals/families (variable expressivity;

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P 26875674

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Modifier genes. Not firmly established for MASS (*data not available*); modifiers of the TGF-β axis are hypothesized.

Epigenetic information. No MASS-specific DNA-methylation/histone signature is established (*data not available*). (Epitranscriptomic A-to-I RNA editing dysregulation has been reported in Marfan aortic smooth muscle,

P 42794678

— *disease-spectrum, not MASS-specific*.)

Chromosomal abnormalities. None characteristic; MASS is a single-nucleotide/small-indel disorder, not a copy-number/aneuploidy syndrome. Large FBN1 deletions are rare.

Ontology: gene FBN1 (HGNC:3603); GO:0005201 extracellular matrix structural constituent; GO:0005520 insulin-like growth factor binding (fibrillin domains); CHEBI:29108 calcium(2+) (cbEGF calcium binding).

5. Environmental Information

- **Environmental factors (toxins/radiation/occupational):** none cause MASS (*not applicable* as etiology). Mechanical wall stress and hypertension aggravate the cardiovascular component.
- **Lifestyle factors:** strenuous isometric/competitive exercise and poorly controlled blood pressure increase aortic/valvular risk; smoking and stimulants are discouraged. Pregnancy is a high-load state warranting cardiovascular surveillance (*human clinical*, P 42880760).
- **Infectious agents:** none (*not applicable*). Standard endocarditis-prophylaxis considerations may apply to significant mitral valve disease per general cardiology guidelines.

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6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A heterozygous **FBN1 variant** (most often a cbEGF-domain cysteine missense; alternatively a null allele) **results in** structurally abnormal or quantitatively reduced **fibrillin-1 monomers**.
2. Abnormal/insufficient fibrillin-1 **leads to** defective **microfibril assembly and reduced extracellular-matrix integrity** in elastic and non-elastic connective tissue (dominant-negative *or* haploinsufficient mechanism) (*review*, P 26875674).
3. Defective microfibrils **result in loss of normal sequestration of latent TGF-β** in the matrix, which **leads to** increased bioavailability and **enhanced TGF-β / Smad2/3 signaling** (*inferred from spectrum data and mouse models*; P 26919284).
4. The chain then **branches**:
5. **4a (Cardiac–valve branch):** enhanced TGF-β signaling + weak valvular matrix **leads to myxomatous degeneration of the mitral valve** → mitral valve prolapse ± regurgitation. "Enhanced TGF-β signaling caused by mutations in Fibrillin-1 (FBN1) ... leads to

myxomatous degeneration of the mitral valve" (*model organism*,

P 42274425

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6. **4b (Cardiac–aortic branch):** weakened aortic medial elastic lamellae + altered smooth-muscle mechanosignaling **result in mild aortic root dilatation**; in MASS this branch is **attenuated/non-progressive** (Z-score <2), whereas in Marfan it progresses to aneurysm/dissection (*human clinical*,

P 20591885

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mechanosignaling e.g. FGF12,

P 41540272

— *disease-spectrum*).

7. **4c (Skin branch):** reduced dermal elastic-fiber integrity **leads to striae atrophicae / skin laxity**.
8. **4d (Skeletal branch):** altered TGF-β/BMP signaling in perichondrium/periosteum **leads to bone overgrowth, pectus deformity, scoliosis, joint laxity** (*inferred from Marfan skeletal biology*).
9. Collectively these tissue-level changes **produce** the clinical MASS tetrad; the *absence* of lens zonule disruption (normal ciliary-zonule fibrillin function in these individuals) **explains the lack of ectopia lentis** that separates MASS from Marfan.

Supporting detail

- **Molecular pathways:** TGF-β receptor signaling (GO:0007179) via Smad2/3; secondary non-canonical ERK/MAPK and AngII–AT1R mechanotransduction (rationale for ARBs;

P 41195532

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Reactome: "Molecules associated with elastic fibres," "Signaling by TGF-beta."

- **Cellular processes:** extracellular-matrix disassembly/remodeling, fibrosis, inflammation (macrophage infiltration in degenerating valves;

P 42274425

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altered vascular smooth-muscle-cell phenotype.

- **Protein dysfunction:** fibrillin-1 misassembly / reduced incorporation into microfibrils; dominant-negative poisoning of multimers; loss-of-function haploinsufficiency.

- **Tissue-damage mechanisms:** elastic-fiber fragmentation and medial changes; myxomatous valvular proteoglycan accumulation.
- **Immune involvement:** low-grade, secondary (macrophage infiltration in valve degeneration; *model organism*, 42274425); not autoimmune.
- **Metabolic/biochemical:** no primary enzyme defect; a structural/signaling ECM disorder. (Metabolic modifiers of aortic disease, e.g., LDHA/lactate, reported in Marfan mice — 42364561; *disease-spectrum*.)

Upstream vs downstream: FBN1 variant (upstream) → microfibril/TGF-β dysregulation (intermediate) → organ phenotypes (downstream). **Cell types (CL):** CL:0000359 vascular associated smooth muscle cell; CL:0002548 fibroblast of cardiac tissue / valve interstitial cell; CL:0000057 fibroblast (dermal); CL:0000136? (not used) — chondrocyte/osteoblast lineages (skeletal). **GO biological processes:** GO:0048251 elastic fiber assembly; GO:0030199 collagen fibril organization; GO:0007179 TGF-β receptor signaling pathway; GO:0030198 extracellular matrix organization. **Subcellular (GO cellular component):** GO:0005578 proteinaceous extracellular matrix; GO:0062023 collagen-containing extracellular matrix; GO:0001527 microfibril.

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7. Anatomical Structures Affected

Organ level (primary): - **Mitral valve** (UBERON:0002135) — myxomatous degeneration, prolapse. - **Aorta / aortic root (sinuses of Valsalva)** (UBERON:0000947; UBERON:0004178 root) — mild, non-progressive dilatation. - **Skin** (UBERON:0002097) — striae, laxity. - **Skeleton / axial & appendicular bones, thoracic cage, vertebral column** (UBERON:0004288 skeleton; UBERON:0003129 skull; UBERON:0001130 vertebral column) — overgrowth, pectus, scoliosis. - **Joints** (UBERON:0000982) — hypermobility.

Body systems: cardiovascular (primary), integumentary, musculoskeletal; ocular (mild myopia only).

Secondary/complication organs: left atrium/ventricle (volume overload from mitral regurgitation); feet (pes planus).

Tissue/cell level: connective tissue (elastic and collagenous ECM) is the unifying target. Cells: vascular smooth muscle cells (CL:0000359), valve interstitial cells/fibroblasts, dermal fibroblasts (CL:0000057), chondrocytes/osteoblasts (skeletal).

Subcellular: extracellular **microfibril** (GO:0001527) and elastic fiber; secretory pathway of fibrillin-1.

Localization / lateralization: systemic and **bilateral/symmetric**; cardiac lesions are central (mitral valve, aortic root).

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8. Temporal Development

- **Onset:** congenital genotype; clinical features typically emerge in **childhood to young adulthood**; insidious, chronic onset.
- **Progression:** generally **stable to slowly progressive**. The defining aortic component is **non-progressive** (Z<2). MVP may progress to significant mitral regurgitation in a subset; scoliosis/pectus can progress during pubertal growth.
- **Disease course / duration: chronic, lifelong.**
- **Patterns / remission:** no spontaneous remission; features are managed, not cured.
- **Critical periods / vulnerability windows:** pubertal growth spurt (skeletal progression); **pregnancy** (cardiovascular load;

  P 42880760

  ); importantly, **childhood evaluations may be inconclusive** — the revised Ghent nosology cautions that mild/young presentations require **serial re-evaluation** because aortic or ocular features may declare later and reclassify MASS as Marfan syndrome (*human clinical*,

  P 20591885

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9. Inheritance and Population

- **Inheritance: autosomal dominant** (HP:0000006). ~50% transmission risk; de novo variants occur.

- **Penetrance:** high but **age-dependent and feature-dependent**; **expressivity is highly variable** even within a family (P 2739055).
- **Epidemiology:** no MASS-specific prevalence figure is established. As a mild FBN1 fibrillinopathy it sits within the broader spectrum; **Marfan syndrome prevalence is ~2–3 per 10,000 births** (~20–30 per 100,000) (*human clinical*, P 21904900), and MASS/overlap phenotypes are plausibly at least as common but under-ascertained (*data not fully available*).
- **Anticipation:** none (not a repeat-expansion disorder).
- **Germline mosaicism:** reported for FBN1 generally; relevant to recurrence counseling (*spectrum data*).
- **Founder effects / consanguinity:** not characteristic (dominant, pan-ethnic); consanguinity not relevant.
- **Carrier frequency:** not applicable (dominant; "carriers" are affected).
- **Population demographics:** pan-ethnic, worldwide; no strong ethnic predilection. **Sex:** both sexes affected; aortic complications across the FBN1 spectrum trend more severe in males, while MVP/mitral regurgitation is somewhat more frequent in females (*human clinical*, P 23622922).

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10. Diagnostics

Diagnostic framework — revised Ghent nosology (2010). MASS is diagnosed when there is MVP, borderline aortic involvement (**aortic root Z-score <2**), systemic connective-tissue features (systemic score typically ≥5), and **no ectopia lentis**, in a patient not meeting Marfan criteria (*human clinical*, P 20591885).

"Diagnosis should be established using the revised Ghent nosology in combination with genotypic analysis in specialized Marfan centres" (P 21904900).

Clinical tests. - *Imaging:* **transthoracic echocardiography** (NCIT:C16818) — aortic root Z-score, MVP, mitral regurgitation; serial surveillance. CT/MRI angiography for complete aortic assessment when indicated. - *Ophthalmology:* **slit-lamp examination with dilated pupils** to exclude ectopia lentis (key discriminator). - *Skeletal assessment:* systemic-score maneuvers (wrist/thumb sign, arm-span/height, pectus, scoliosis, pes planus, facial features), spine radiographs as needed. - *Laboratory:* **no diagnostic blood/urine biomarker** exists; homocysteine/urine amino acids to exclude homocystinoria in differential. - *Pathology (if valve resected):* myxomatous proteoglycan-rich degeneration.

Genetic testing. FBN1 molecular analysis is central: single-gene sequencing + deletion/duplication analysis, or (preferred) a **multigene aortopathy/connective-tissue NGS panel** covering FBN1, TGFBR1, TGFBR2, SMAD3, TGFB2, TGFB3, ACTA2, MYH11, COL3A1, etc., to resolve the differential. WES/WGS are useful when panels are negative. Chromosomal microarray/karyotype/FISH/mtDNA/repeat testing are **not indicated** (not a CNV/mitochondrial/repeat disorder). NCIT:C16858 Genetic Testing.

Omics-based diagnostics: not used clinically for MASS (*not applicable*).

Clinical criteria: revised Ghent nosology (2010) is the standard (P 20591885).

Differential diagnosis: - **Marfan syndrome** — ectopia lentis and/or progressive aortic root aneurysm (Z≥2) present. - **Loeys–Dietz syndrome** — arterial tortuosity, hypertelorism, bifid uvula; TGFBR1/2, SMAD3, TGFB2/3. - **Ehlers–Danlos syndromes** — skin/joint predominance; vascular EDS (COL3A1). - **Isolated/familial MVP** and **familial ectopia lentis** (allelic FBN1). - **Homocystinuria** — lens dislocation + thrombosis + biochemical markers.

Screening: no newborn/population screening. **Cascade clinical + genetic screening of first-degree relatives** is recommended once a familial FBN1 variant is identified.

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11. Outcome / Prognosis

- **Survival / life expectancy:** generally **favorable and near-normal**, because MASS **lacks the progressive aortic aneurysm** that drives mortality in Marfan syndrome — where "aortic disease is the main cause of death" and, before surgery, "most patients died in the fourth decade of life" (*human clinical*,

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P 20351702

MASS mortality is not elevated by the defining phenotype.

- **Disease-specific mortality:** low; no MASS-specific mortality registry (*data not available*). Risk would rise only if a patient is actually on the Marfan end of the spectrum (hence surveillance).
- **Morbidity / function:** mainly musculoskeletal (pain, scoliosis, pectus, joint instability), significant mitral regurgitation in a minority (→ arrhythmia, heart failure, rarely need for valve repair/replacement), and skin changes. Disability is usually mild.
- **Complications:** progressive mitral regurgitation; development/declaration of aortic dilatation over time (reclassification); pregnancy-related cardiovascular risk.
- **Prognostic factors:** degree and progression of aortic root dimension and mitral regurgitation; genotype (null vs dominant-negative) and family history of dissection; age and sex.
- **Prognostic biomarkers:** none validated specifically for MASS; serial aortic Z-score and echo are the practical predictors (*human clinical*).
- **QoL measures:** no MASS-specific EQ-5D/SF-36/PROMIS data (*data not available*).

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12. Treatment

There is **no cure and no disease-specific drug**; management targets the cardiovascular component and symptomatic skeletal issues, mirroring fibrillinopathy care.

Pharmacotherapy (cardiovascular, when aortic dilatation/MVP warrants): - **Beta-blockers** (e.g., atenolol) — reduce hemodynamic wall stress/aortic growth. NCIT:C2667 (Beta-Blocker); atenolol NCIT:C236. - **Angiotensin II receptor blockers** (losartan) — attenuate TGF-β/AT1R-driven aortic pathology; NCIT:C61876 (Losartan). In Marfan RCTs, "Aortic root diameter increased with no differences between groups: 0.4 mm/year" for losartan vs atenolol — i.e., **comparable aortic protection** (*human clinical RCT*,

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P 30261963

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P 23622922

Inverse-agonist ARB activity is mechanistically important (*model organism*,).

P 41195532

- *Pharmacogenomics:* no FBN1-specific PGx guidance (*not applicable*).

Advanced therapeutics: gene therapy, cell therapy, RNA-based therapy, immunotherapy — none approved or standard; not applicable to MASS currently (*data not available*). Allele-specific/ASO approaches are conceptual for dominant-negative FBN1.

Surgical/interventional: rarely needed for MASS; **mitral valve repair/replacement** for severe mitral regurgitation; prophylactic aortic root surgery is generally **not required** in true MASS (reserved for Marfan-range aortic dilatation; valve-sparing root replacement/David procedure or Bentall when applicable —).

P 21904900

Supportive/rehabilitative: physical therapy for scoliosis/joint instability; orthotics for pes planus; orthopedic management of pectus/scoliosis; pain management; activity counseling (avoid isometric/competitive sports with significant cardiac involvement).

Experimental: FBN1-spectrum trials (e.g., ARB/beta-blocker comparisons; emerging TGF-β and metabolic targets) are Marfan-focused (*disease-spectrum*, PMIDs 30261963, 42364561). No MASS-specific NCT trials identified (*data not available*).

Treatment strategy: individualized, severity-based; combination beta-blocker + ARB is used in some aortic cases; genotype-guided risk stratification and multidisciplinary (cardiology, genetics, orthopedics, ophthalmology) care.

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13. Prevention

- **Primary prevention:** not possible (germline genetic cause). **Preimplantation genetic testing (PGT-M)** and **prenatal diagnosis** can prevent transmission when a familial FBN1 variant is known.
- **Secondary prevention: serial echocardiographic surveillance** of the aortic root and mitral valve for early detection of progression/reclassification; ophthalmologic monitoring.
- **Tertiary prevention:** beta-blocker/ARB therapy, blood-pressure control, activity modification, endocarditis-risk awareness, and pregnancy planning to prevent

complications (*human clinical*,

P 42880760

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- **Immunization / public-health / environmental / infectious interventions:** not applicable.
- **Genetic counseling:** essential — autosomal-dominant 50% recurrence risk, variable expressivity, importance of cascade testing and longitudinal re-evaluation; per ACMG/NSGC/GeneReviews frameworks.
- **Risk stratification:** identify relatives carrying the familial variant for targeted surveillance.

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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** human FBN1 (Gene 2200); mouse **Fbn1** (NCBI Gene 14118; NCBI Taxon **10090**, *Mus musculus*); rat Fbn1 (Taxon 10116); zebrafish *fbn1* (Taxon 7955). Fibrillin-1 and microfibril biology are **evolutionarily conserved** across vertebrates.
- **Natural disease in other species:** fibrillinopathies/elastic-fiber disorders occur in animals; a **bovine FBN1-associated connective-tissue/"Marfan-like" syndrome** is catalogued in OMIA (fibrillin-1 cattle). MASS as a named entity is human-specific; comparable overlap phenotypes are not formally defined in veterinary medicine (*data limited*).
- **Comparative pathology:** animal FBN1 disruption reproduces aortic/valvular and elastic-fiber pathology, supporting conserved mechanism.
- **Transmission / zoonosis:** not applicable (genetic disorder).

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15. Model Organisms

- **Principal model — mouse.** The heterozygous knock-in **Fbn1^C1039G/+** (cysteine substitution in a cbEGF domain — the same variant *class* seen in patients) is the standard FBN1-fibrillinopathy model. It recapitulates **myxomatous mitral valve degeneration/prolapse**, **progressive thoracic aortic aneurysm**, and **elastic-fiber fragmentation**, driven by **enhanced TGF-β signaling** (*model organism*,

P 42274425

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Secondary elastic-tissue phenotypes (e.g., lacrimal-gland dysfunction) also appear
( ). P 41533936

- **Other genetic models:** hypomorphic/null Fbn1 alleles (e.g., mgR, mgΔ) model haploinsufficiency/severe disease; conditional and transgenic lines exist.
- **Model types:** mammalian in vivo (mouse, rat); zebrafish *fbn1*; patient-derived fibroblasts/iPSC-derived smooth-muscle and valve cells (in vitro).
- **Translational use / applications:** these models established the **TGF-β–antagonism rationale**: "treatment with losartan normalized aortic root growth and aortic wall architecture" in an FBN1-targeted mouse (*model organism*,
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directly motivating human ARB trials. They are used to dissect mechanosignaling (FGF12;
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immune/lymphatic contributions to valve degeneration
( ), P 42274425
and metabolic modifiers
( ). P 42364561
- **Model limitations (for MASS specifically):** existing mice model the **Marfan (progressive-aortic) end** of the spectrum; they do **not** cleanly reproduce the *mild, non-progressive aortic* MASS phenotype, so MASS-specific genotype–phenotype questions (why some FBN1 alleles yield only MASS) remain **under-modeled**.
- **Resources:** MGI (Fbn1), IMPC/IMSR, Alliance of Genome Resources, ZFIN.

Evidence Summary (key PMIDs)

PMID	Type	Contribution
2739055	Human clinical	Original MASS definition; phenotypic continuum
20591885	Human clinical (criteria)	Revised Ghent nosology; MASS vs Marfan discriminators
26875674	Human clinical/ review	FBN1/fibrillin-1 biology; type-1 fibrillinopathy spectrum; R2726W mild allele
26919284	Review	FBN1→TGF-β pathway; therapeutic shift to TGF-β antagonism
42274425	Model organism	FBN1→TGF-β→myxomatous mitral valve degeneration
42222959	Model organism	Standard Marfan mouse aortic model
17892982	Model organism/ clinical rationale	Losartan normalizes aortic growth in FBN1 mouse
30261963 / 23622922	Human clinical RCT	Losartan vs atenolol aortic outcomes
21904900	Human clinical	Prevalence 2–3/10,000; combined Ghent+genotype diagnosis; aortic surgery
20351702	Human clinical	Marfan mortality driven by aortic disease
42880760	Human clinical	Pregnancy cardiovascular risk in FBN1 disorders
41195532 / 41540272 / 42364561	Model organism	ARB mechanism; mechanosignaling; metabolic modifiers (spectrum)

Limitations and Future Directions

1. **MASS-specific data are sparse;** much evidence is extrapolated from Marfan/FBN1-spectrum studies (explicitly flagged). Dedicated MASS natural-history cohorts, prevalence, QoL, and mortality data are needed.
2. **Genotype–phenotype determinants** of why specific FBN1 alleles yield only the mild MASS phenotype are unresolved; modifier genes are uncharacterized.

3. **No MASS-specific model** reproduces the non-progressive aortic phenotype.
4. **Diagnostic caution:** MASS in children/young adults may evolve toward Marfan; longitudinal re-evaluation is essential.
